First Non-Repeating Character • Find the first non-repeating character in a string using HashMap.

eclipse-workspace - Day_8_Java_Assignments/src/com/assignments/day8/FirstUniqueChar.java - Eclipse IDE

```java
package com.assignments.day8;
import java.util.*;
public class FirstUniqueChar {
    public static void main(String[] args) {
        String input = "simplification";

        // Map to store Character and its Frequency
        Map<Character, Integer> charCountMap = new HashMap<>();

        // Step 1: Count occurrences of each character
        for (char c : input.toCharArray()) {
            charCountMap.put(c, charCountMap.getOrDefault(c, 0) + 1);
        }

        // Step 2: Find the first character with a count of 1
        boolean found = false;
        for (char c : input.toCharArray()) {
            if (charCountMap.get(c) == 1) {
                System.out.println("Input String: " + input);
                System.out.println("First non-repeating character is: " + c);
                found = true;
                break;
            }
        }

        if (!found) {
            System.out.println("No unique character found.");
        }
    }
}
```

```
<terminated> FirstUniqueChar [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (30 Apr 2026, 10:55:03 am – 10:55:03 am elapsed 0:00:00) [pid: 4556]
Input String: simplification
First non-repeating character is: s
```

Online Shopping Cart • Use List for cart, Map for product-price mapping.

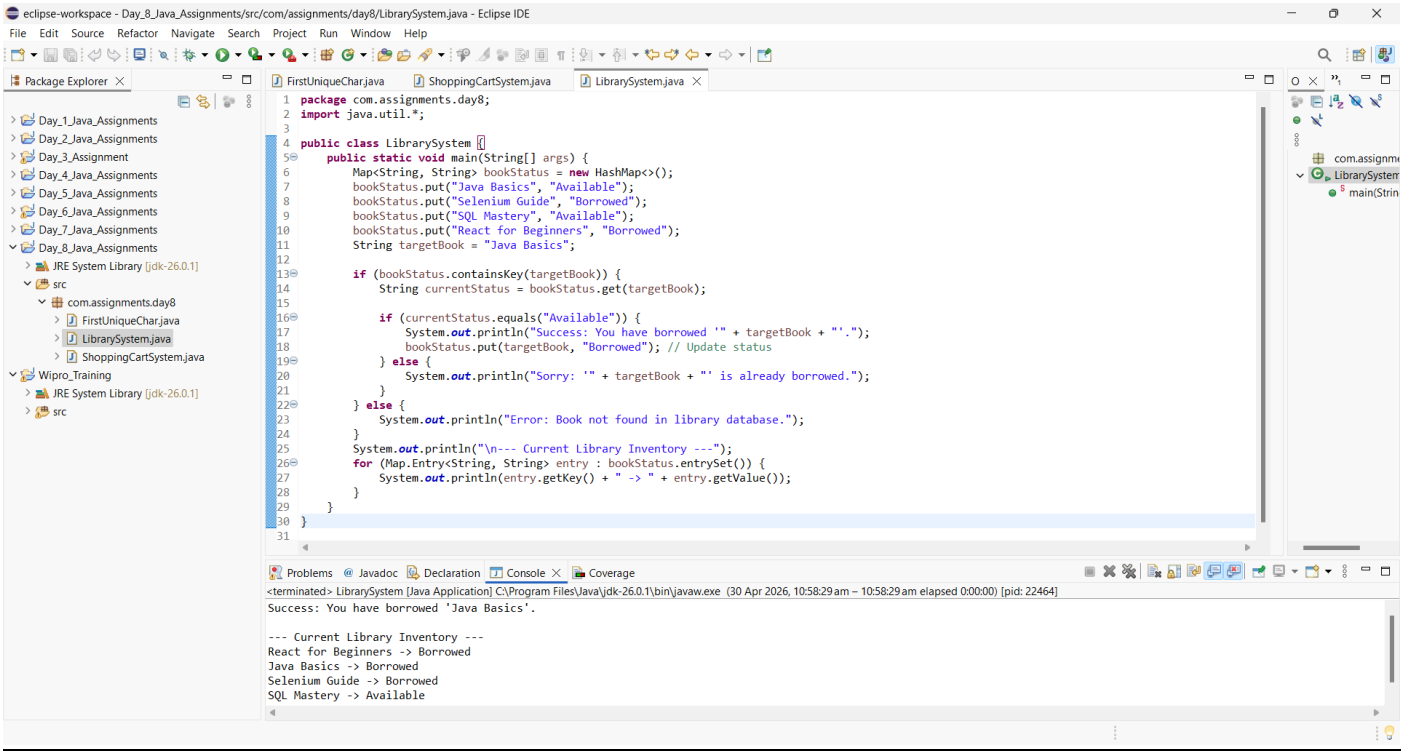
eclipse-workspace - Day_8_Java_Assignments/src/com/assignments/day8/ShoppingCartSystem.java - Eclipse IDE

```java
package com.assignments.day8;
import java.util.*;
public class ShoppingCartSystem {
    public static void main(String[] args) {
        Map<String, Double> priceList = new HashMap<>();
        priceList.put("Laptop", 45000.0);
        priceList.put("Mouse", 500.0);
        priceList.put("Keyboard", 1200.0);
        priceList.put("Monitor", 8000.0);

        List<String> myCart = new ArrayList<>();
        myCart.add("Laptop");
        myCart.add("Mouse");
        myCart.add("Mouse");
        myCart.add("Keyboard");

        double totalBill = 0;
        System.out.println("--- Your Receipt ---");

        for (String item : myCart) {
            if (priceList.containsKey(item)) {
                double price = priceList.get(item);
                System.out.println(item + " : " + price);
                totalBill += price;
            }
        }
        System.out.println("--------------------");
        System.out.println("Total Amount: " + totalBill);
    }
}
```

```
<terminated> ShoppingCartSystem [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (30 Apr 2026, 10:58:49 am – 10:58:49 am elapsed 0:00:00) [pid: 10216]
--- Your Receipt ---
Laptop : 45000.0
Mouse : 500.0
Mouse : 500.0
Keyboard : 1200.0
--------------------
Total Amount: 47200.0
```

Library Management • Use Map to track availability.